Training and Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Training and Testing before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Training and Testing trace and record the first step you can explain without help.

Use W2: Distinguish frames from **group**s for Splits and related examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Training and Testing explanation against the supplied facts before discussing it.

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Training and Testing target to a trace and a justified conclusion.

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased **unseen** test result. Explain the issue and revise the evaluation plan.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Training and Testing case and show the deciding evidence.

A rule is developed and evaluated using the same books A–H.

Why does this not establish **unseen** performance?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test **evidence**?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Training and Testing adds to Transparent Book Recommender and keep the evidence.

Keep final user profiles hidden until weights and thresholds are fixed.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased **unseen** test result. Explain the issue and revise the evaluation plan.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the three roles

Explain the three roles. Apply the supplied Training and Testing facts and keep your reasoning.

Explain what **training**, validation and **test set**s are used for. What changes if you repeatedly choose a model using its test score?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Distinguish frames from groups

> Distinguish frames from groups. Apply the supplied Training and Testing facts and keep your reasoning.

For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

My reasoning and evidence

W3 Repair a related example leak

> Repair a related example leak. Apply the supplied Training and Testing facts and keep your reasoning.

A1 and A2 are in **training** and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

My reasoning and evidence

W4 Audit and repair independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit and repair independently. Apply the supplied Training and Testing facts and keep your reasoning.

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

My reasoning and evidence

W5 Match separation to the claim

> Match separation to the claim. Apply the supplied Training and Testing facts and keep your reasoning.

A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative.

Is this enough for a strong performance claim?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Training and Testing

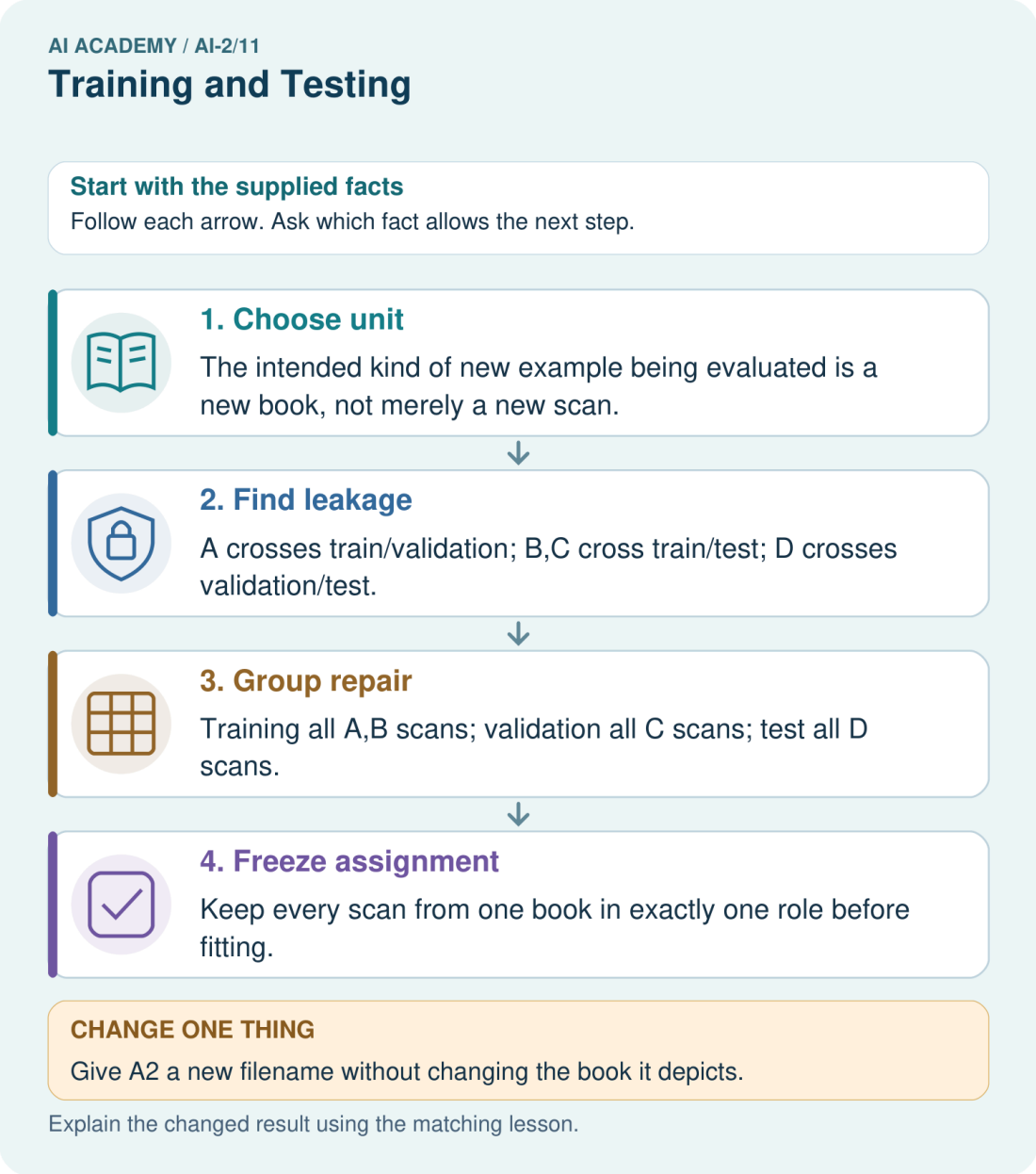
> A fictional catalogue model learns from scans of book covers and will face previously unseen

> books. Explain the labelled step before checking the changed condition.

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.



Follow the labelled picture with the matching explanation. Open the vector diagram

How many distinct books are in each repaired role, and why report that as well as scan counts?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: **Group** the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep every scan from one book in exactly one role before **fitting**.

1. **Group** repair

2. Choose unit

3. Find **leakage**

4. Freeze assignment

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Freeze assignment Keep every scan from one book in exactly one role before **fitting**. Different scan IDs do not make examples independent books.

A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give A2 a new filename without changing the book it depicts. Which conclusion is supported by the supplied facts?

1. Its **group** is still A; the original related examples still cross the split.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its **group** is still A; the original related examples still cross the split. Renaming a row does not remove the relationship between scans.

Repair a missing building block

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

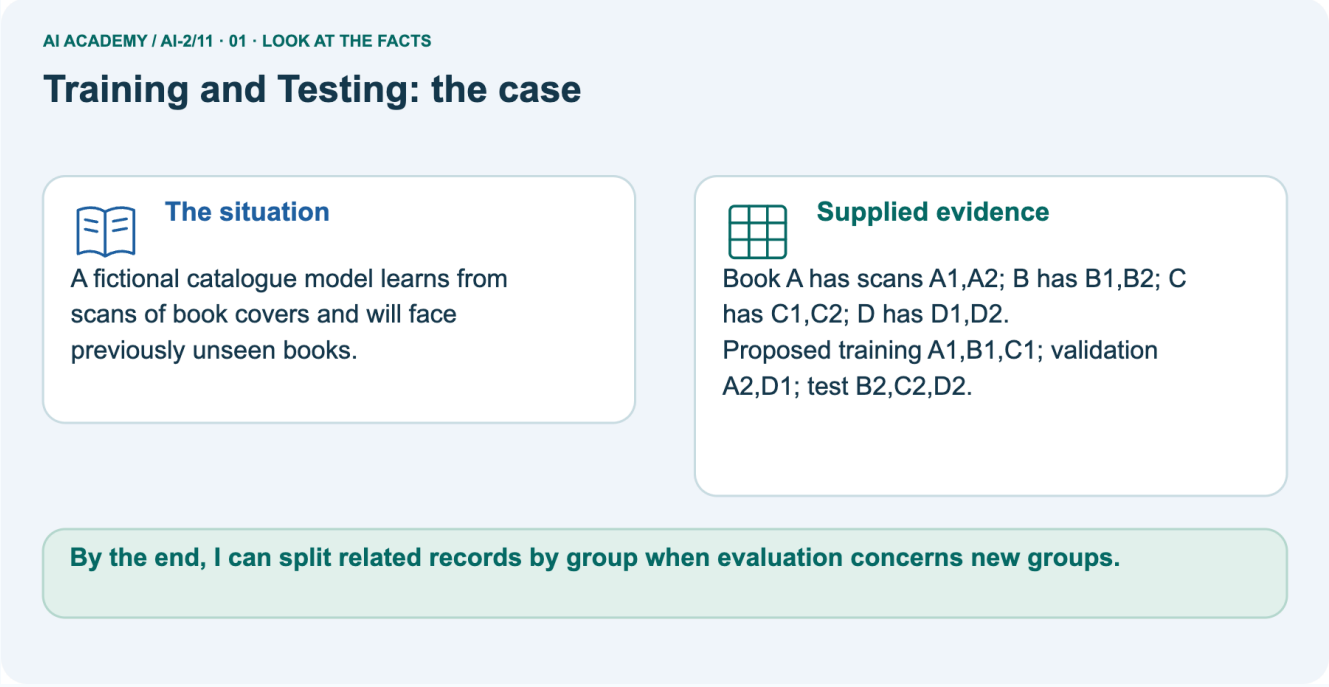
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Training and Testing picture from its labels, then record a relevant limit.

01. Training and Testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

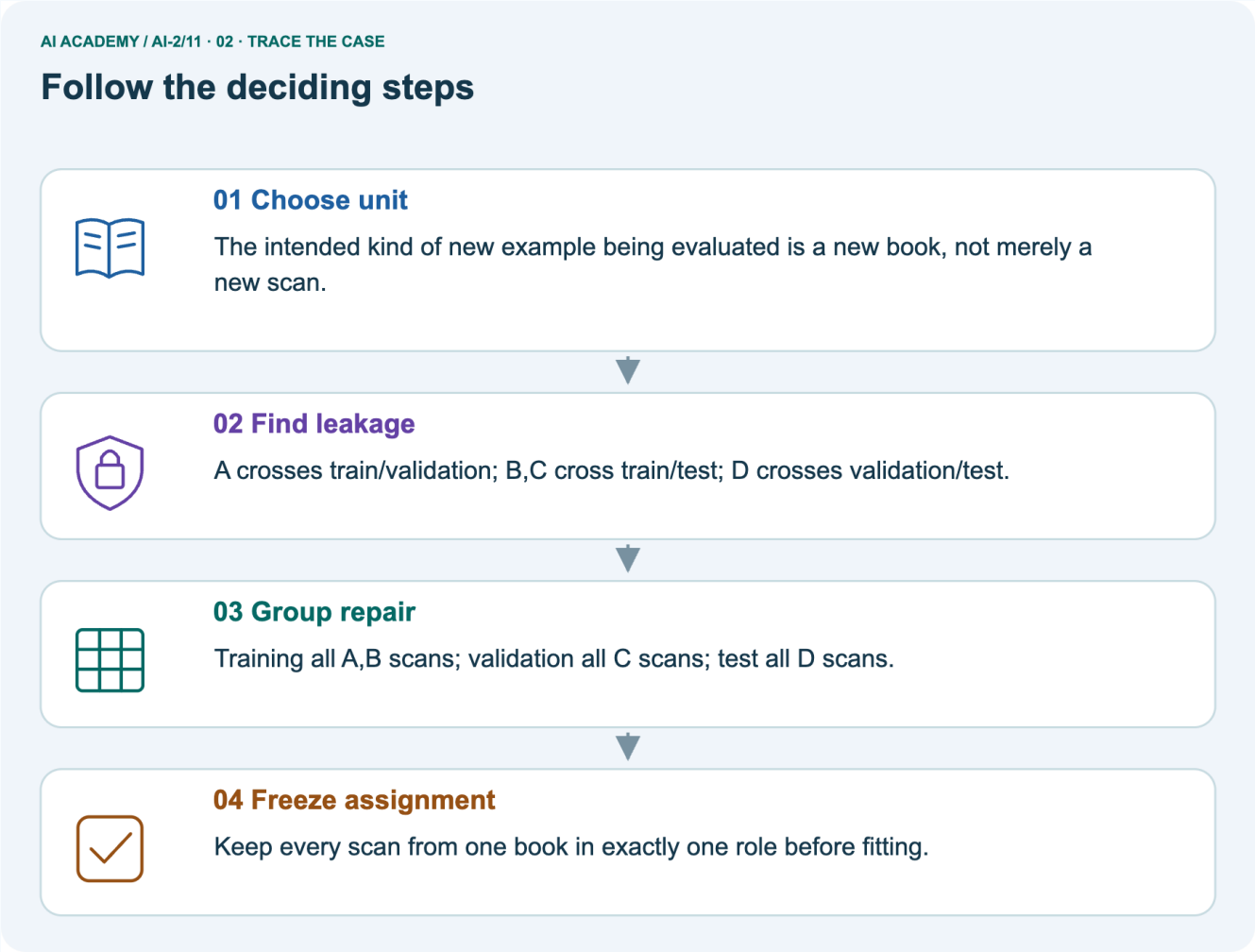
A fictional catalogue model learns from scans of book covers and will face previously **unseen** books. Supplied facts: Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



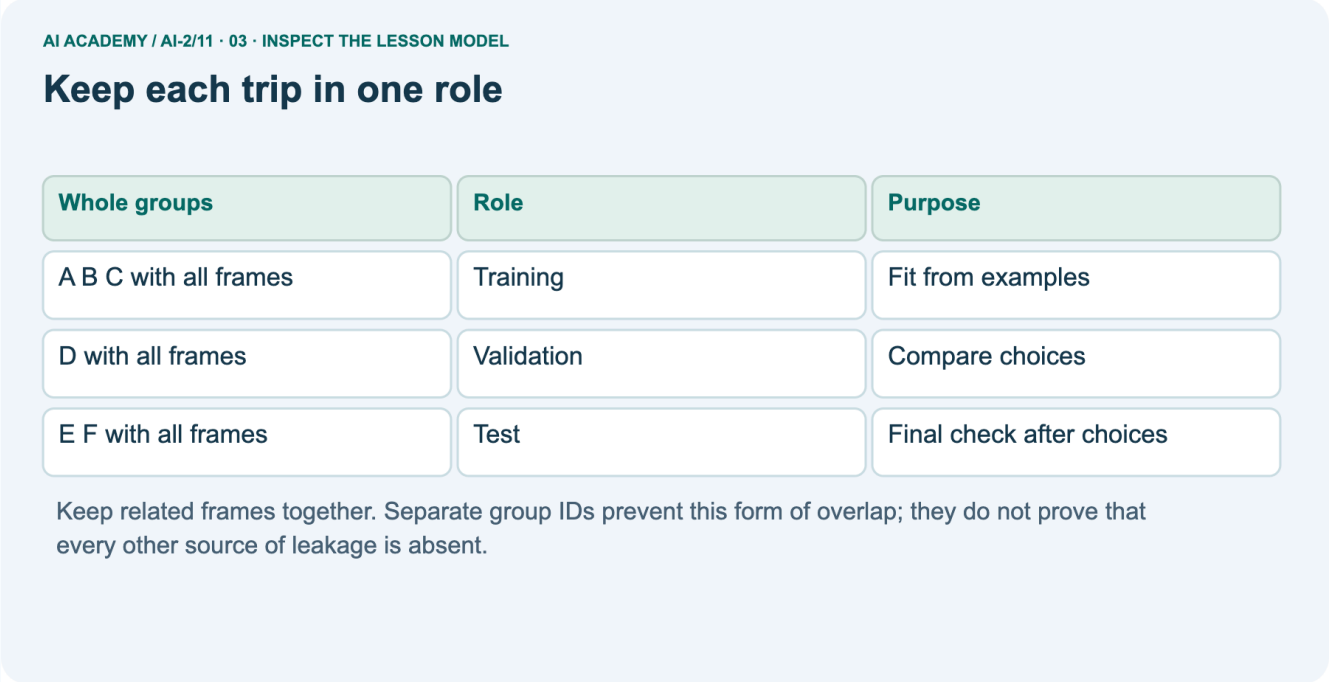
Follow the labelled picture with the matching explanation. Open the vector diagram

Choose unit: The intended kind of new example being evaluated is a new book, not merely a new scan. → Find **leakage**: A crosses train/validation; B,C cross train/test; D crosses validation/test. → **Group** repair: **Training** all A,B scans; validation all C scans; test all D scans. → Freeze assignment: Keep every scan from one book in exactly one role before **fitting**.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep each trip in one role



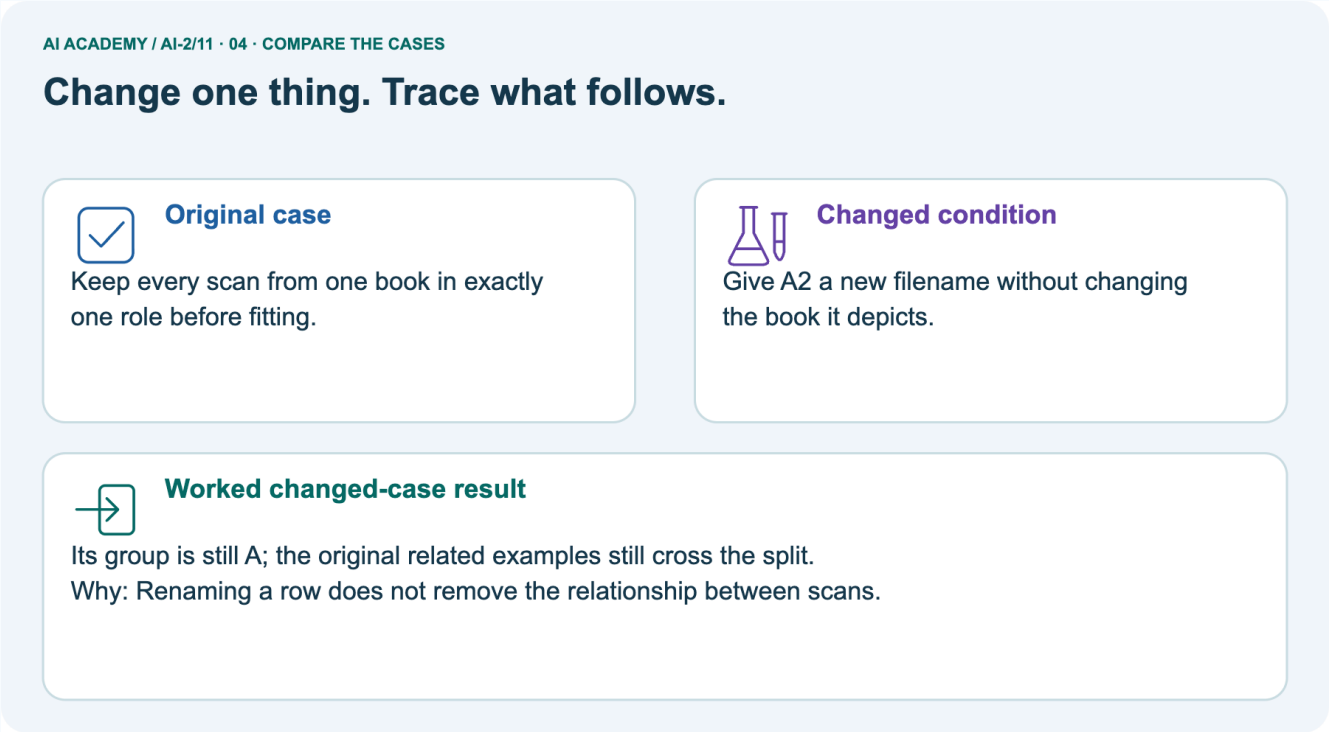
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep each trip in one role. A B C with all frames | **Training** | Fit from examples. D with all frames | Validation | Compare choices. E F with all frames | Test | Final check after choices

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

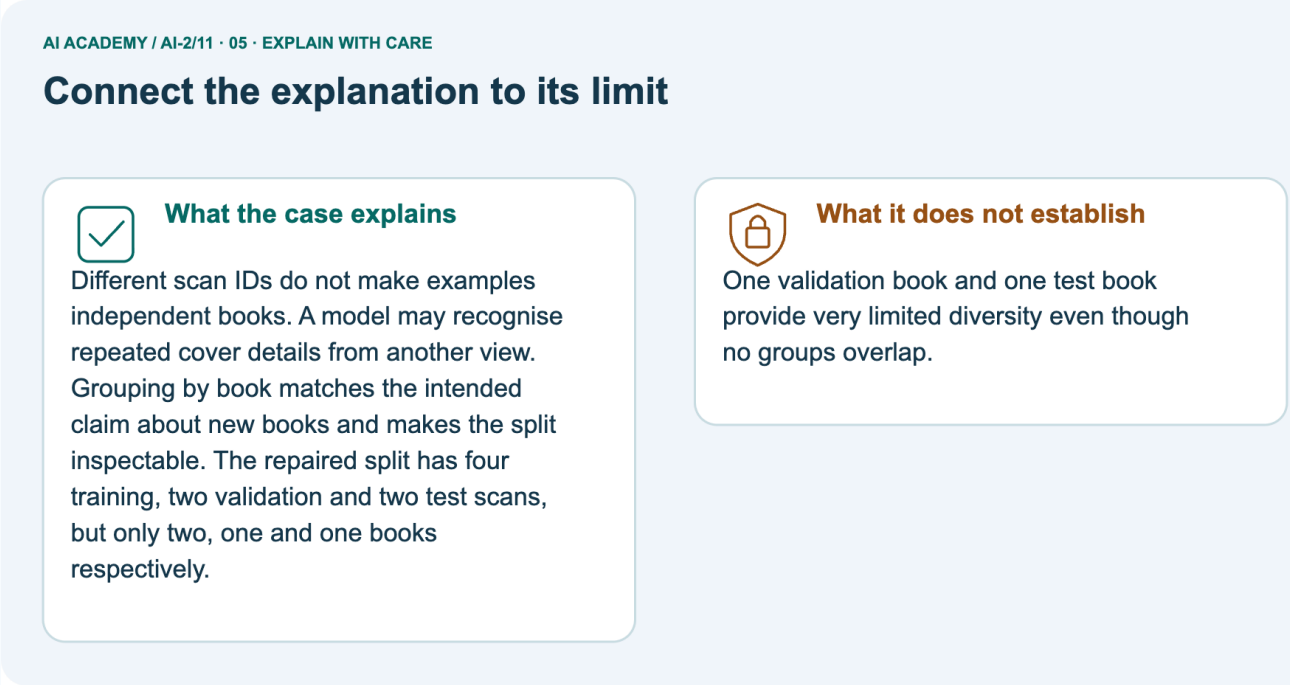
Original case: Keep every scan from one book in exactly one role before **fitting**. Changed condition: Give A2 a new filename without changing the book it depicts.

Expected result: Its **group** is still A; the original related examples still cross the split. Reason: Renaming a row does not remove the relationship between scans.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

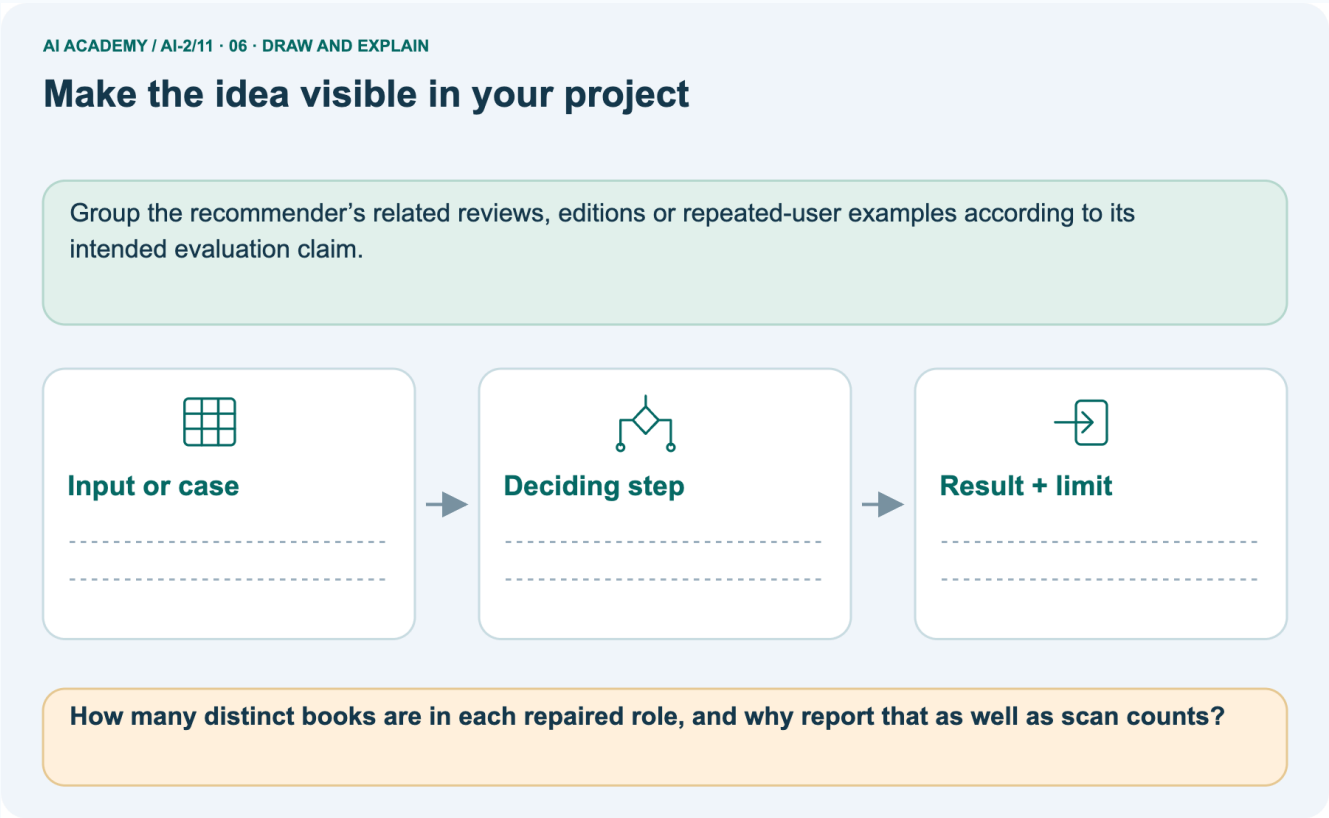
Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively. Limit: One validation book and one test book provide very limited diversity even though no **group**s overlap.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim. Draw your **input**, deciding step, result and limit.

Supported question: How many distinct books are in each repaired role, and why report that as well as scan counts?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.